SECTOR BRIEFING | JULY 2026

Robots on the Battlefield, Farm, and Doorstep

How autonomous systems are reshaping defence, food production, and the home

Coverage	Defence & Unmanned Systems, Precision Agriculture, Consumer & Eldercare Robotics
Focus	Applications, enabling technology, market economics, and adoption outlook
Prepared for	Robotics Applications Study
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Executive Summary

Service robots – defined by the International Federation of Robotics (IFR) as machines that perform useful tasks for people or equipment outside of pure factory manufacturing – are among the fastest-growing segments of global robotics. Unlike industrial arms fixed to a factory floor, service robots work in unstructured, changing settings alongside people, animals, and rough terrain.

This briefing moves from the highest-stakes domain to the most familiar: it begins with **military** unmanned systems, continues through **agricultural** robotics, and closes with **domestic** service robots, before turning to shared economic dynamics and the outlook ahead.

The core conclusion is straightforward: service robotics is shifting from novelty device to mission-critical infrastructure. Defence forces increasingly rely on unmanned systems as a standing element of doctrine; precision-agriculture robots are answering labour shortages and sustainability pressure; and household robots have normalised autonomous machines in daily life. Shared enabling technology – sensing, machine learning, edge computing, and better batteries – is pulling all three domains together and speeding up cross-sector innovation.

Reading the sectors in this order also makes the underlying pattern easier to see: the same handful of technology building blocks recur again and again, tuned to very different levels of risk, supervision, and regulatory scrutiny. A sensor package built to survive electronic warfare and one built to map a living room draw on the same lineage of advances in perception and autonomy software, even though the consequences of failure are worlds apart.

KEY POINTS

- Three domains covered: military, agricultural, and domestic service robotics
- Common technology base: sensing, AI/ML, edge computing, battery systems
- Notable business-model shift: Robotics-as-a-Service (RaaS)
- Central tension: expanding autonomy versus regulation, safety, and ethics

1 Comparative Analysis at a Glance

Before examining each domain in depth, the table below summarises the defining characteristics of all three, presented up front as a reference point for the sector-by-sector discussion that follows.

Dimension	Military	Agricultural	Domestic
Primary goal	Risk reduction & readiness	Yield & sustainability	Convenience & elder care
Environment	Extreme, adversarial	Outdoor, variable	Indoor, structured
Key sensors	Ruggedised vision, radar, EW-resistant comms	RTK-GPS, multispectral imaging	LiDAR / vision SLAM
Autonomy level	Variable, human-in-loop for force	High, moderate stakes	High, low stakes
Adoption barrier	Cost, ethics, regulation	Capital cost, connectivity	Cost, privacy

2 Introduction

The global service-robotics market has expanded quickly over the past decade, propelled by advances in artificial intelligence, sensor miniaturisation, and falling hardware costs. Per the IFR, service robots split broadly into two categories: personal or domestic-use robots, and professional-use robots spanning agriculture, defence, logistics, healthcare, and hospitality. This report covers three domains – ordered here as military, agricultural, and domestic – that together illustrate how broadly service robots now touch national security, food production, and everyday life.

Each domain carries distinct technical demands:

- **Military** systems must withstand extreme environments and resist jamming or cyberattack, operating where failure has severe consequences.
- **Agricultural** systems must endure harsh outdoor conditions, cover large areas with minimal supervision, and interoperate with existing farm machinery.
- **Domestic** systems must be affordable, safe around untrained users, and able to work in cluttered, ever-changing home environments.

Taken together, these three requirement sets frame the rest of the report. The following sections walk through each domain's applications and enabling technologies in turn, before a shared discussion of economic impact and the trends likely to shape adoption over the next decade.

RISK REDUCTION & READINESS

Military Service Robots

2.1 Overview

Military and defence robotics span unmanned ground vehicles (UGVs), unmanned aerial vehicles (UAVs), unmanned maritime and underwater systems, and explosive-ordnance-disposal

(EOD) robots. These platforms lower risk to personnel during reconnaissance, logistics, and hazardous-material handling while extending operational reach and endurance.

2.2 Key Application Areas

- Reconnaissance and surveillance UAVs delivering real-time battlefield awareness
- EOD robots – tracked platforms that inspect and neutralise suspected explosives
- Logistics and resupply UGVs moving equipment across difficult terrain
- Unmanned maritime vehicles for mine countermeasures, harbour security, and undersea surveillance
- Robotic exoskeletons augmenting soldier strength and endurance

2.3 Technology and Considerations

Military-grade robots require ruggedised hardware, secure jam-resistant communications, and a degree of autonomy that can function under GPS-denied or electronic-warfare conditions. The use of lethal autonomous weapon systems raises active ethical and legal debate; most current deployments keep a human operator in or on the decision loop for use-of-force actions. International bodies, including the United Nations, continue discussing frameworks to govern increasingly autonomous weapons, reflecting broad concern over accountability and unintended escalation.

Beyond the ethical dimension, procurement decisions in this domain are shaped by durability and lifecycle cost as much as raw capability: a platform that must operate for years in harsh terrain, survive repeated redeployment, and resist tampering or reverse-engineering if captured places very different demands on materials and software architecture than a consumer device ever would.

YIELD & SUSTAINABILITY

Agricultural Service Robots

2.4 Overview and Drivers

Agricultural robotics – precision agriculture, or agri-robotics – is driven by persistent farm-labour shortages, rising input costs, and demand for sustainable, chemical-efficient farming. Robots now handle seeding, weeding, spraying, harvesting, and livestock monitoring at a precision manual labour cannot match.

2.5 Representative Applications

- Autonomous tractors and seed-drills on GPS-guided paths with centimetre-level accuracy
- Robotic weeders using machine vision to separate crops from weeds and apply targeted treatment
- Fruit-picking robots with soft grippers and vision systems that judge ripeness while avoiding bruising
- Autonomous drones for crop-health monitoring, multispectral imaging, and targeted spraying

- Robotic milking and livestock-monitoring systems that track animal health

2.6 Technology Enablers and Impact

These systems rely heavily on RTK-GPS positioning, multispectral and hyperspectral imaging, and machine-learning models trained to spot crop stress, pests, and optimal harvest timing. Benefits include lower pesticide and water use, reduced labour dependency, more consistent yields, and rich field data for long-term soil and crop management. Barriers include high upfront capital cost, the need for reliable rural connectivity, and the complexity of operating across variable field conditions and crop types. Smallholder farms in developing economies face an added barrier: fragmented, small-plot land holdings that make large autonomous machinery uneconomical, favouring lighter, modular tools and shared-ownership cooperative models instead.

The data generated in the course of routine operation is itself becoming a secondary product of agricultural robotics: soil composition readings, yield maps, and pest-pressure histories accumulated season after season give operators a longitudinal picture of field performance that was previously impossible to compile at this level of granularity or cost.

CONVENIENCE & ELDER CARE

Domestic Service Robots

2.7 Overview and Market Adoption

Domestic service robots are the most visible, widely adopted category, led by robotic vacuum and floor-washing robots from makers such as iRobot, Ecovacs, Roborock, and Xiaomi. These devices use LiDAR or visual SLAM to map the home for systematic cleaning rather than random movement. The category also includes lawn-mowing robots, robotic pool cleaners, companion and eldercare robots, and smart-home hubs with simple robotic arms.

2.8 Core Technologies

- SLAM and computer vision for indoor mapping and obstacle avoidance
- Object recognition to navigate around furniture, cables, and pets
- Cloud connectivity for scheduling, firmware updates, and voice-assistant integration
- Rechargeable lithium-ion batteries with automatic docking and self-charging

Eldercare and companion robots add fall detection, voice interaction, and health-vital tracking, often linked to family members or care providers through mobile apps.

2.9 Benefits and Challenges

The main benefit is time saved and improved quality of life, especially for elderly or disabled users. Challenges include a price premium on high-end models, privacy concerns from continuous home-mapping and cameras, and terrain and battery limits such as stairs, thick carpet, and thresholds. As populations age across many industrialised economies, eldercare and companion robots are expected to grow fastest in this category, aided by government-funded pilots addressing shortages of professional caregivers.

The privacy question deserves particular attention as this sub-category matures: continuous in-home sensing that is acceptable, even reassuring, when used for fall detection or health monitoring raises very different concerns when the same data streams could be repurposed, shared, or exposed through a security lapse – a tension manufacturers and regulators are still working through.

3 Economic and Market Impact

The economic footprint of service robotics has grown substantially as unit costs fall and reliability improves.

DOMESTIC

Mass production and competition among Asian and North American manufacturers have pushed entry-level robotic-vacuum prices down while adding capability, widening the addressable consumer base each year.

AGRICULTURAL

The business case is increasingly driven by labour economics: seasonal farm labour has grown scarcer and costlier in many regions, making autonomous machinery a direct substitute rather than a luxury upgrade – particularly for high-value, labour-intensive crops such as fruits and vegetables.

MILITARY

Procurement of unmanned systems has become a standing budget line for most modern armed forces, reflecting a doctrinal shift toward moving risk away from personnel and toward expendable or reusable machines.

3.1 Robotics-as-a-Service (RaaS)

A second economic effect is the rise of RaaS business models, particularly in agriculture and, increasingly, eldercare. Instead of buying hardware outright, farms and care facilities can subscribe to a bundled service covering the robot, software updates, maintenance, and data analytics as a predictable operating expense. This lowers the capital barrier and shifts performance risk to the manufacturer – a model expected to speed adoption among small and mid-sized operators who would otherwise be priced out of automation.

4 Future Outlook

Convergence defines the trend across all three domains. Gains in battery energy density, edge-AI processing, and low-cost LiDAR are benefiting home robots, farm equipment, and defence platforms at the same time. Robot fleets coordinated through cloud-based fleet-management software are expected to spread further across agriculture and logistics-oriented military use. Regulatory frameworks covering autonomy, safety certification, and – in the military context – the ethical use of autonomous force will increasingly shape the pace and direction of adoption over the next decade.

5 Conclusion

Service robots have moved firmly past experimental status into practical, revenue-generating, mission-critical use across defence forces, farms, and homes. Each domain carries its own technical and regulatory demands, but all three share a common technology base and a shared trajectory toward greater autonomy, intelligence, and fleet-level coordination. Organisations and policymakers that invest early in supporting infrastructure, workforce training, and governance frameworks will be best placed to capture the productivity and safety gains these systems offer.

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